Aafthab Ali Shaik

Software Engineer

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Professional Summary

I am a Computer Science Student and I have experience in building applications that can handle a lot of users and machine learning systems. I know a lot about the basics of Computer Science, like data structures and algorithms. I also know how to build the backend of a website make APIs and design databases. I have built applications that're ready to be used by people using React.js, Node.js and TensorFlow. I want to be a Software Engineering Intern to help make systems that work well and can handle a lot of users.

Technical Skills

Programming: C, Python, Java, C Sharp
Frontend: React.js, TypeScript, JavaScript, Vite
Backend: Node.js, Express.js, Flask, FastAPI, REST APIs
Databases: MySQL, PostgreSQL, MongoDB, Firebase, Supabase
Cloud & Tools: Git, GitHub, GitLab CI/CD, GitLab Runner, Docker, Render, Vercel, Hugging Face Transformers.js
Testing: Pytest, Playwright, Vitest, React Testing Library
Core Concepts: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Static Code Analysis, Embedded Software Build Automation

Experience

•Defence Research and Development Organisation (DRDO) Research Centre Imarat (RCI) *May 2026 – Jul 2026*
Software Engineering Intern Hyderabad

– Designed and developed a CI/CD pipeline for embedded software build, verification, and static analysis using GitLab CI/CD.
– Configured GitLab Runners and Docker containers to automate embedded software build workflows.
– Performed static code analysis and generated automated build reports to improve software quality.
– Collaborated with DRDO scientists on embedded software development and DevOps workflow optimization.

•Centre for Extended Reality (CXR), GITAM University *May 2026 – Jun 2026*
Full Stack Developer Intern Visakhapatnam

– Developed and enhanced the **CXR Web Portfolio** by improving both frontend and backend user interfaces for a responsive and user-friendly experience.
– Integrated Firebase services by implementing API calls for data retrieval, storage, authentication, and real-time updates while ensuring seamless frontend-backend communication.
– Refactored the codebase by fixing bugs, improving code quality, resolving security vulnerabilities, and optimizing application performance for a stable production-ready deployment.

•Suvehi Interactive Technologies Pvt. Ltd. *Jan 2026 – Apr 2026*
Software Engineering Intern Visakhapatnam

– Designed and developed an internal company website by implementing responsive frontend interfaces and scalable backend functionalities.
– Worked on multiple software development projects, including website development, game version upgrades, feature implementation, software maintenance, and bug fixing.

•Infosys Springboard *Dec 2025 – Feb 2026*
Full Stack Developer Intern

– Designed and implemented a full-stack web application using React.js, Node.js, and MySQL.
– Developed RESTful APIs and optimized backend request handling.
– Implemented JWT-based authentication and integrated Chart.js for financial data visualization in TaxPal project.

•BharatVersity *Apr 2025 – Jul 2025*
AI/ML Intern BITS Hyderabad

– Built a Music Recommendation System using Retrieval-Augmented Generation (RAG).
– Implemented data preprocessing, feature engineering, and model evaluation pipelines.
– Automated ML workflows and structured data pipelines for improved retrieval accuracy.

Projects

•AI Voice Assistant Dec 2025

Developed an AI-powered voice assistant integrating OpenAI and Gemini APIs with real-time speech processing and response generation.

•TaxPal Personal Finance & Tax Estimator Feb 2026

Developed a full-stack web application using React.js, Node.js, and MySQL with RESTful APIs for managing income, expenses, and tax estimation. Implemented JWT-based authentication for secure user access and integrated Chart.js for visualizing financial insights through interactive dashboards.

•Real-Time Facial Expression Detection Web App Sep 2026

Built and deployed a real-time facial expression detection web application using React.js, Flask, ONNX Runtime, and MediaPipe. Implemented live camera-based emotion detection, Android/iOS browser compatibility fixes, GitHub Actions testing, Docker deployment, and Render keep-alive health checks.

–Live Project Link
–GitHub Repository Link

•RouteWise – Smart Journey Planner with Learning Fares Sep 2026

Built and deployed a real-time journey planner for Visakhapatnam using React, TypeScript, FastAPI, Google Maps APIs, and Supabase PostgreSQL. Implemented multi-modal route comparison for buses, autos, cabs, bikes, and walking based on estimated cost and travel time, along with a Bayesian fare model using rider fare votes, outlier rejection, and time-decayed weighting. Added live GPS navigation with off-route detection, Google/Apple Maps hand-off, Docker deployment on Render, and 62 automated tests.

–Live Project Link
–GitHub Repository Link

•StudyMind – AI Study Workspace & Interview Coach Sep 2026

Built and deployed an AI study platform using React.js, Vite, Firebase, and the Google Gemini API. Implemented PDF parsing with in-browser OCR (Tesseract.js) and hybrid BM25 + MiniLM retrieval for page-cited answers, generating notes, 2/5/8-mark question banks, and document-grounded quizzes. Built an Interview Studio with adaptive Gemini follow-ups and on-device speech recognition (Whisper via Transformers.js). Added an admin dashboard, Firestore security rules, 10x faster uploads, 160 Vitest tests, and Render deployment.

–Live Project Link
–GitHub Repository Link

•SortScope – Interactive DSA Visualizer & Code Tracer Oct 2026

Built and deployed a DSA learning lab using JavaScript, Vite, Node.js, and WebAssembly. Implemented reversible step-by-step playback for five sorting algorithms with synchronized pseudocode, shared-input comparisons, and 70 operations across seven data structures. Built a Code Lab that traces Python (Pyodide) and JavaScript programs line by line in the browser, visualizing variables, call stacks, and tree/graph diagrams. Added 29 Vitest and 30 Playwright tests and Vercel deployment; next, adding C, C++, and Java via a rate-limited Oracle Cloud compiler API.

–Live Project Link
–GitHub Repository Link

Education

•GITAM University *Aug 2023 – Present*

B.Tech in Computer Science / CGPA: 8.06 Visakhapatnam

Certifications

•Microsoft Azure AI Fundamentals (AI-900) Apr 2024

Score: 781

Leadership

- Core Member Google Developers Group (GDG), GITAM University 2025 – Present